# Nick Mazurkin — Software Engineer, Team Lead, Staff Engineer

**New York City, USA**, greencard, +13477203170, career@mazurk.in, LinkedIn, GitHub, HF, Resume

**Java**, **Python**, **Web**, **DB**, **No-SQL**, **Linux**, **ML**, **AI**, **Cloud**, **K8S**

Software development professional with years of experience crafting web applications and backend services using Java and Python. I am skilled in search and machine learning technologies, web technologies, application architecture, database design, and Linux system administration.

## Experience

**PulsePoint, New York, USA**

PulsePoint.com, AdTech, OpenRTB Exchange, DSP, DMP

Java · Spring Framework · Project Reactor · Lucene · JUnit · Jetty · Netty · Disruptor · Ant · Maven · Linux · Kubernetes · Docker · Graphite · Graphana · Kibana · Prometheus · Git · Python · Jupyter · Jupyter · NumPy · Scikit-Learn · Luigi · ETL Redis · Cassandra · Hive · No-SQL · Presto · Vertica · AWS S3 · AWS EC · AWS SageMaker · Google Cloud BigQuery · Google Cloud Storage · Algorithms · Algorithm Design · Data Structures · Machine Learning Algorithms · Real-Time Bidding (RTB) · Multi-Threading · Concurrency

**VP, Data Science, ML and Engineering team (Mar 2025 — PRESENT)**
**Director, Data Science, ML and Engineering team (Apr 2020 — Mar 2025)**
**Senior Software Engineer with Expertise in Java, Python, and ML (Jan 2019 — PRESENT)**
**Senior Software Engineer with Expertise in Java, Python, and ML (remote contractor, Feb 2017 - Jan 2019)**

Led a team of 6 data scientists and 3 Java/Python engineers (DSE) working across 5 time zones, responsible for planning, performance tracking, coordination, and agile management. Proven ability to lead and mentor teams, manage projects, and deliver results on time and within budget.

- Successfully delivered 3 new internal services and 5 new ML models as a team leader.
- Successfully completed 20 quarterly plans with the team in research, development, and deployment.
- Successfully migrated all DSE/ML projects to the new data center in 3 months.
- Provided interviewing, on-boarding, and mentoring for new team members.
- Successfully coordinated and communicated with other teams and team leaders on numerous projects.
- Successfully maintained an iterative Agile framework, including scrum, retrospectives, and prioritization of tasks.

Experienced software engineer with a proven track record of developing and deploying high-performance applications using Java, Python, ML, Linux, and web technologies. Expertise in building and scaling distributed systems, implementing machine learning models, and optimizing performance for large-scale data processing.

- Developed an asynchronous web crawler/fetcher for /ads.txt and /app-ads.txt web resources, providing and enhancing transparency and trust for the SSP and DSP partners of our Exchange platform (processing ~500K domains daily, Java, Spring, Netty).
- Developed a supply prediction system for RTB traffic, enabling clients to switch to a cheaper self-serving platform (processing 24M documents daily using Lucene, Spring Framework and ML/CatBoost).
- Implemented a scoring model to effectively throttle proxy processing of incoming supply, selecting only 20% of the most profitable RTB requests and blocking 80% of the least profitable supply. This resulted in a 50% reduction in the number of required servers.
- Designed and implemented a DMP platform using asynchronous Java, gRPC, Cassandra, and Redis. This platform provides the effective solutions for targeting and optimization algorithms, accessing 30 billion entities with a p99 latency of 5ms and a p50 latency of 1ms.
- Developed a standalone service for domain attribution, enabling efficient domain and application bundle targeting (achieving 100k/sec QPS using gRPC and REST).
- Developed a client-side balancing library using the JSQ algorithm and circuit-breakers, significantly reducing client-server communication latency from p95=6ms to p95=1ms.
- Containerized and deployed a cluster of DoubleVerify service instances, achieving 300k/sec QPS in peak. Implemented custom client-side balancing and a low-latency custom client-to-cluster connection HTTP/REST library, enabling faster and easier deployments with efficient access.
- Developed a dockerized test infrastructure for custom configuration of Redis and Cassandra, similar to TestContainers, which facilitated the execution of integration tests.
- Developed a hierarchy of docker images for ~20 projects written in Python, Java, and Jupyter, significantly reducing the time required to create new projects.
- Developed an audience builder for 15 audiences, enabling our client to target very specific audiences with focus either on volume or accuracy (using Luigi, Hive, and ETL).
- Successfully resolved 2-3 support issues daily as part of a periodical 3rd-line technical on-call support rotation.
- Implemented several dozen RTB changes in the application stack, including improvements, new features, deployments, fixes, and monitoring (using Java, Netty, and Linux).
- Developed an engine for internal reports with embedded alerting, enabling transparent and quick monitoring of important pipelines and services (using Python, Hive, Impala, and Jupyter). Created 6 internal engineering reports.
- Implemented 7 ETL pipelines in Python using Luigi, Docker, Hive, and Presto.
- Improved video-completion rate model by making it more accurate and stable (Python, Scikit-learn)
- Implemented and delivered to production the look-alike audience model (Python, Catboost, Luigi, K8S)
- Developed a model to predict the quality of NPI attribution, enabling clients to select the confidence level for HCP targeting.
- Developed a model to predict the quality of ICD10 attribution based on obfuscated and aggregated clean-room medical data prescription data, enabling clients to select DTC audiences based on condition probability.
- Developed a secure facade to access and process obfuscated clean-room data, providing a secure and efficient interface to

private data (Python, Flask, Gunicorn, Google Cloud API).
- Implemented and delivered the gender prediction model (Bayes, SQL, Python, Sklearn)
- Implemented the experimental win-rate model for the avails predictor (Python, PyTorch)
- Implemented and delivered the versioned ML model storage (Nginx, Python, Cherrypy, Docker, K8S)
- Implemented and delivered the app store scraper with embeddings (Python, Docker, K8S, Trino, PyTorch, Optuna, Tensorboard)
- Implemented and delivered the cleanroom data visualizer (Python, Streamlit, Plotly, Google BigQuery, Google Big Storage)
- Implemented and delivered the NPI embedding model for the HCP (Python, Docker, PyTorch, Contrastive Learning, Google Big Query, Google Big Storage)
- Implemented and delivered the domain embedding model (Python, Docker, PyTorch, HDBSCAN, Google BigQuery, Google Big Storage)
- Implemented and delivered the semantic encoder with medical/health specialization (Python, PyTorch, Lightning, Hugging Face, multi-GPU)
- Implemented and the AI/LLM "Council" project to build/improve ML model automatically base on the selected metrics (Python, ACP, Copilot CLI)
- Implemented and delivered the medical contextualizer (Python, PyTorch, Lightning, HF, Transformers, FastAPI, Docker, K8S)
- Implemented and delivered the medical entity extractor (Python, PyTorch, GLiNER, HF, Transformers, UMLS, FastAPI, Docker, K8S)
- Implemented and delivered the new video completion rate model (Python, Java, Catboost, Google BigQuery, Docker, K8S)

## Deutsche Bank

Deutsche Bank Technology Center, Moscow, RU

**Senior Software Java Engineer, AVP (Jun 2016 — Feb 2017)**

Java · Linux · Spring Framework · Angular · Solace · KDB

- Implemented several new algorithms, code and concurrency improvements to the dbCAP project achieving the required performance of the solution (exchange events messaging on Solace System message broker, KDB+, java).
- Successfully implemented the test suite for the legacy projects allowing to extend the life-time and reliability of the legacy projects (process control, runtime log analyzers, TeamCity, environment settings).
- Improved the internal messaging framework based on Solace Systems message broker (achieved up to 160000 messages/sec stored reliably to the disk), introduced solutions for the performance metering.
- Designed and implemented the multithreaded and garbage-less algorithms and structures (Java) for message acknowledgement in complex environments to provide effective processing without long GC pauses.
- Implemented a modern front-end build toolchain for the legacy project, added the required UI features for the project (Angular 2, NPM, RxJs, TypeScript, TSLint, System.js, Rollup, jQuery, Bootstrap).

## OK.ru

OK.ru, Moscow, RU, the one of the largest social networks (54 MAU)

**Senior Software Java Developer (Oct 2012 — May 2016)**

Java · Linux · Lucene · Cassandra · Kafka · Spring Framework · jQuery · ML · LVS

- Implemented a multiple features in the searching infrastructure of the OK.ru service: backend services, data storages, relevancy customization, indexation, replication (Apache Lucene, Spring Framework, Web Services, SQL/noSQL, Linux)
- Fixed multiple legacy issues in L12N infrastructure for the OK.ru service including the resources editor, Java frameworks and workflow process, achieved resource propagation to ~2000 servers.
- Make significant contribution to the moderation and antispam services, architecture and implementation, implemented NSFW detector, spam detector (based on naive bayes detector), UGC filter cluster (near real-time voting-quorum service).
- Successfully designed, implemented and run the MVP for the startup project (Tinder-alike) in very short time (3 months), created the architecture of the service including SOA, custom search service, asynchronous binary client protocol, push service (GCM), photo uploading/processing/distribution service, statistics and analytics (apache netty, apache cassandra, kafka, queues, messaging, mobile apps, python, machine learning).

## HeadHunter Group

HeadHunter.ru, The largest job portal in Russia, Moscow, RU

**Team lead, Software Developer (Jun 2011 — Oct 2012)**

Java · Linux · PostgreSQL · RabbitMQ · Maven · Spring Framework · Struts · XSLT · JSP · MemCached · Hibernate · Python

- coordination of technical aspects of team work (5-6 members)
- resource planning, requirement management, processing in agile and with scrum
- release management (DEB and JAR artifacts, GIT branch control, GIT flow)
- a lot of Java coding
- hands on bugs and production issues

## Luxoft/UBS

Luxoft for UBS investment banks, Moscow, RU

**Senior software developer (Jun 2010 — Jun 2011)**

Java · Linux · ZSH · Tibco · JMX · JMS · SQL · Oracle DB

Data distribution and messaging system (RDDH) for UBS Bank

- A lot of Java code created for RDDH messaging project
- Database stored procedures, DB DDL and meta-data
- Cluster management (ZSH scripts): architecture and implementation
- Integration test suite was implemented for some legacy systems and components
- Worked in the decentralized team (Moscow, Russia - London, UK)

### Internet Lotto Ltd

The social network, Yoshkar-Ola, RU

**Team lead, Software developer (Sep 2008 — Jun 2010)**

Java · J2EE · Linux · Nginx · Apache HTTPd · Apache Tomcat · Apache ActiveMQ · Terracotta · MySQL · Spring Core · Spring WS · Spring WebFlow · Spring Modules · Spring AOP · Sphinx · EhCache · Quartz · iBatis · Velocity · JSP&JSTL · JMX · DWR · JUnit · EasyMock · Apache Lucene/Solr · Apache Ant · JScript · jQuery · jQuery UI · Amazon S3 · Jetty · ImageMagick · MEncoder

- Architecture and implementation of a social network for poker players
- Coding
- Database design and administration (MySQL)
- Transparent integration with payment system moneta.ru
- Integration with poker rooms (statistics, promo links)
- Team management (2 members)

### Moneta.ru

Moneta.ru, Yoshkar-Ola, RU, the online financial service

**Software developer (Mar 2007 — Sep 2008)**

Payment/billing system «moneta.ru»

Java · J2EE · ORACLE & PL/SQL · Spring Core · Spring WebFlow · XSLT

- Database schema design and implementation
- Stored procedures implementation and meta-data management
- DAO layer implementation
- Unit and integration test suites

### Chromatec SCB LLC

Chromatec.ru, Yoshkar-Ola, RU, the gas chromatography equipment and software manufacturer

**Team lead, Software Developer, Electronics Engineer (1996 — 2007)**

Delphi · Borland C/C++ · MS C/C++ · XML/XSLT · Firebird SQL Server · MS Access · MS Visual Basic · Win32 API · CVS · ASM · i8051 · PCAD · OrCAD

- Desktop software for chemical industry working with hardware devices (Delphi, Borland C/C++, MS C/C++, XML/XSLT, Firebird SQL Server, MS Access, MS Visual Basic, Win32 API)
- Digital electronic systems on i8051 architecture (schemes, printboards, software)
- Real-time embedded i8051-software with C and Assembler.

## Open Source projects

### net-crusher (2016)

TCP/UDP proxy library for automated and manual connectivity testing (used by RabbitMQ, Aerospike, Terracotta), more: project, sources, usages

### protobuf (2016)

Performance fix for the protobuf java compiler: pull request

## Education

### Mari State Technical University (1994 — 1999)

- Yoshkar-Ola, Russia, “Computing machines, complexes, systems, and networks”
- Engineer (with honors, equivalent to a master’s degree)
- Graduation work: “Applying of wavelet transformation to results of gas-chromatography analysis”

### Mari State University (2002 — 2003)

- Yoshkar-Ola, Russia, «Anti-Crisis Management»
- Anti-Crisis Manager
- Graduation work: “Staff motivation methods at Chromatec LLC”

### Additional

- 2026, Retrieval Augmented Generation (RAG), Coursera
- 2024, Generative AI for Everyone, Coursera
- 2024, Generative AI with Large Language Models, Coursera
- 2020, Become a Manager, LinkedIn
- 2016, Codility Golden Award (Vanadium 2016), Codility
- 2016, Algorithms: Design and Analysis, Part 2, Coursera
- 2016, Algorithms: Design and Analysis, Part 1, Coursera
- 2016, Economics for non-economists, Coursera
- 2016, Game Theory, Coursera
- 2016, Machine Learning, Coursera
- 2016, An introduction to machine learning, Coursera

## Resume

The latest and the most detailed resume can be downloaded in different document formats at https://mazurk.in